

DWAO Accessibility Widget v2.0.0

Architecture & Security Review — dwao-a11y-widget.js · 2248 lines · Zero dependencies

Architecture Layers

LAYER 1

Vanilla JS & CSS

- IIFE wrapper — no global leakage
- Read `data-*` attrs from `<script>` tag
- Resolve theme / brand color → `BRAND`
- Inject CSS into `<head>`
- Apply position variant CSS

LAYER 2

Data & State

- Inline SVG icon map (`ICONS{}`)
- Section / card config (`accessibilitySections{}`)
- Feature dispatch map (`optionActions{}`)
- Module-level closure state flags
- CSS class name arrays (font, line, letter, align)

LAYER 3

UI Construction

- `createEl()` DOM utility
- `svgIcon()` icon builder
- `createOptionCard()` — 24 feature cards
- `createAccordionSection()` — 4 sections
- `buildAccessibilityPanel()`
- `buildScreenReaderPanel()`

LAYER 4

Feature Engine

- Typography (font size, letter/line spacing, align, dyslexia)
- Visual / Color (contrast, invert, grayscale, saturation)
- Navigation (cursor, links, buttons, skip link, reading line)
- Screen Reader (Web Speech API — hover-to-speak)
- Text Magnifier (floating lens overlay)
- Media Control (audio/video/YouTube/GIF pause)
- Page Structure (heading nav panel)
- Virtual Keyboard (draggable, injects into active input)

LAYER 5

Event & Interaction

- Accordion expand/collapse
- Card click → `optionActions` dispatch
- Keyboard nav (↑↓ Arrow, Enter, Space, Escape)
- Click-outside to close panel
- ARIA live region announcements
- Auto-label unlabelled form fields
- Reset All → `resetAllAccessibility()`

LAYER 6

Persistence

- Key: `accessibility_local_settings`
- Value: JSON array of `datakey` strings active
- Write: on every card click (via `requestIdleCallback`)
- Read: on init — replays clicks to restore state
- Clear: on Reset All

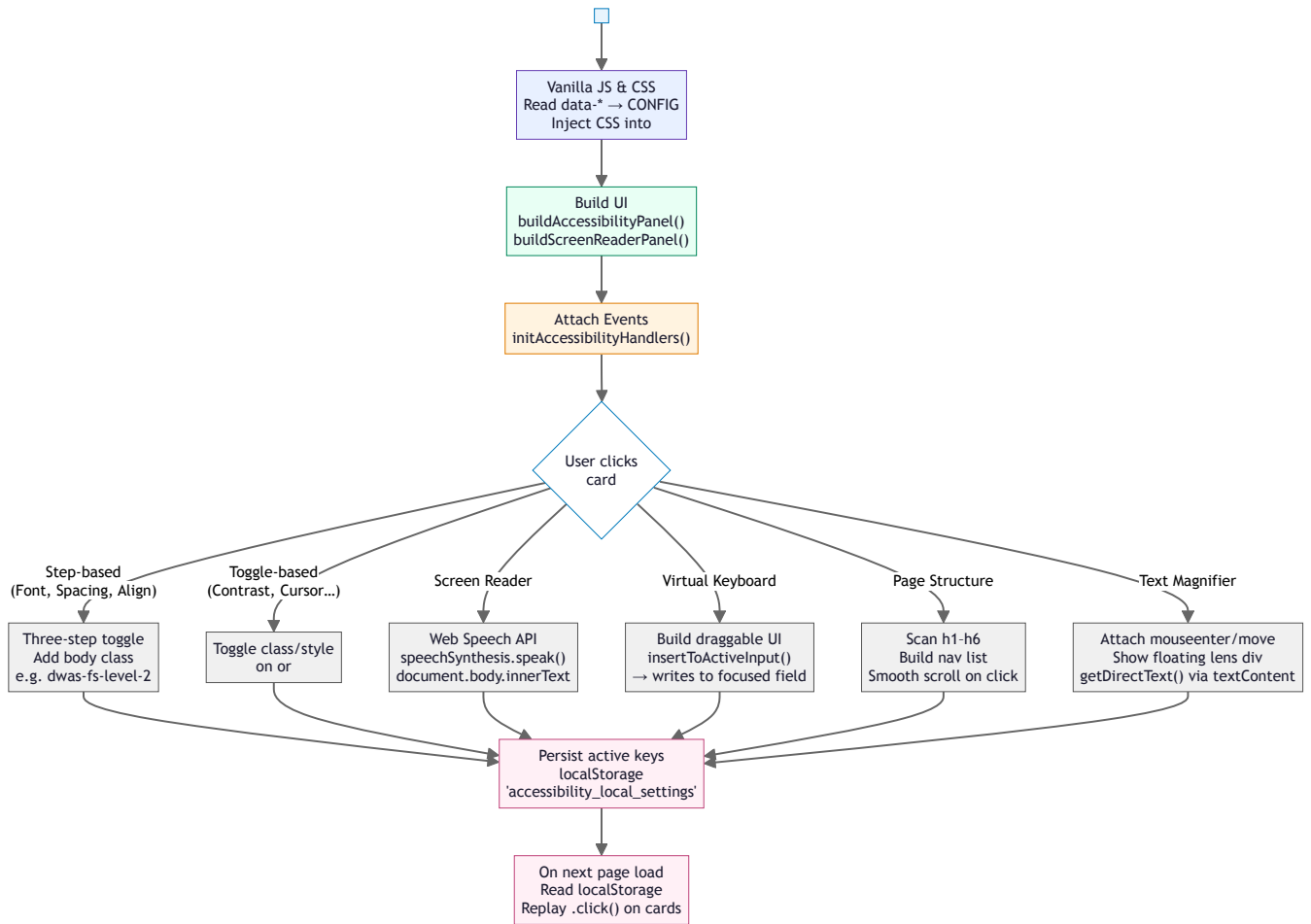
PUBLIC API

window.DWAOAccessibility

- `.version` — "2.0.0"
- `.reset()` — programmatic reset all
- Everything else stays inside IIFE closure

DWAO Accessibility Widget v2.0.0 — Data Flow Diagram

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Feature Map — 24 Accessibility Controls

PROFILES

Blindness

Visually Impaired

Cognitive

Epilepsy

Dyslexia

ADHD

CONTENT

Text Size

Hide/Show Images

Line Spacing

Letter Spacing

Text Zoom

Image Description

Skip Link

Pause Media

Text Alignment

COLOR & CONTRAST

High Contrast

Invert Colors

Grayscale

Low Saturation

NAVIGATION

Reading Line

Highlight Links

Big Cursor

Enlarge Buttons

Read Page

Page Structure

Virtual Keyboard

Security Findings

SEVERITY	FINDING	LOCATION	DETAIL	FIX
HIGH	XSS via <code>img.alt</code>	<code>toggleAltText()</code> Line 1224	<code>tooltip.innerHTML = img.alt</code> — if any image has a crafted <code>alt</code> attribute (e.g. via CMS), HTML executes	Replace with <code>tooltip.textContent = img.alt</code>
MEDIUM	<code>getPageStructure()</code> Line 1756	Halts JS execution when DevTools is open; functional regression + exposes internals		
LOW	Global namespace pollution	Multiple locations	<code>window._dwaoMouseOver</code> , <code>window._magnifierListeners</code> ,	Move into IIFE closure or namespace under <code>window.DWA0Accessibility._internal</code>

SEVERITY	FINDING	LOCATION	DETAIL	FIX
			<code>window.magnifierEnabled</code> leak onto <code>window</code>	
LOW	Brand color CSS injection	Vanilla JS & CSS Lines 186–190	<code>data-brand-color</code> is directly interpolated into CSS strings without validation	Validate value matches a safe CSS color pattern (hex/rgb/named) before use
LOW	localStorage replay via <code>.click()</code>	<code>initAccessibilityHandlers</code> Line 2229	Persisted state is restored by simulating click events — including <code>speechSynthesis.speak()</code> , which requires a real user gesture	Call feature functions directly instead of simulating clicks
INFO	<code>innerHTML</code> footgun in <code>createEl()</code>	<code>createEl()</code> Line 128	All callers currently pass static strings; no active XSS path, but any future caller passing user data is unsafe by default	Add a comment/lint rule; consider a separate <code>createElSafe()</code> that only accepts <code>textContent</code>